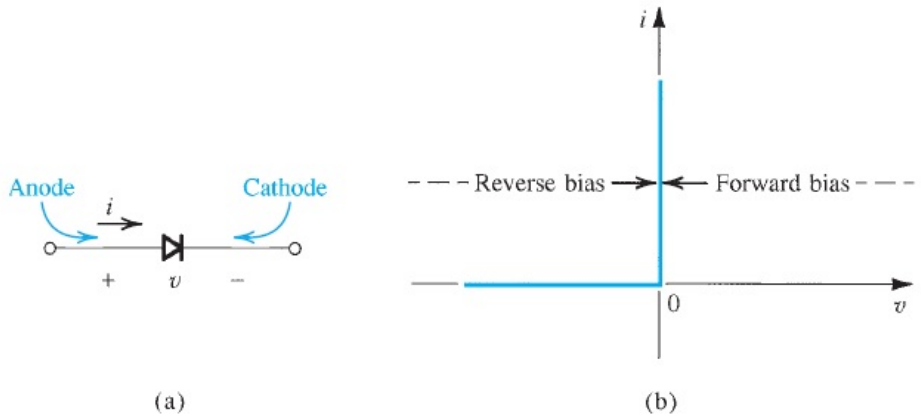
Diodes

- In the scope of this course, we will have a brief introduction to two electronic circuit components: diodes and OPAMPs.
- The simplest and most fundamental nonlinear circuit element is the diode. Just like a resistor, the diode has two terminals; but unlike the resistor, which has a linear (straight-line) relationship between the current flowing through it and the voltage appearing across it, the diode has a nonlinear i–v characteristic.
- Of the many applications of diodes, their use in the design of rectifiers (which convert AC to DC) is the most common.

The Ideal Diode

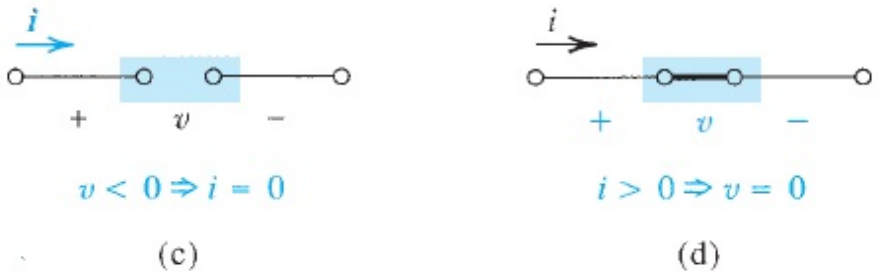
- The ideal diode may be considered to be the most fundamental nonlinear circuit element. It is a two-terminal device having the circuit symbol and the i–v characteristic shown in the figures below.



- If a negative voltage is applied to the diode, no current flows and the diode behaves as an open circuit. Diodes operated in this mode are said to be reverse biased, or operated in the reverse direction. An ideal diode has zero current when operated in the reverse direction and is said to be cut off, or simply off.

The Ideal Diode

- On the other hand, if a positive current is applied to the ideal diode, zero voltage drop appears across the diode. In other words, the ideal diode behaves as a short circuit in the forward direction; it passes any current with zero voltage drop. A forward-biased diode is said to be turned on, or simply on.

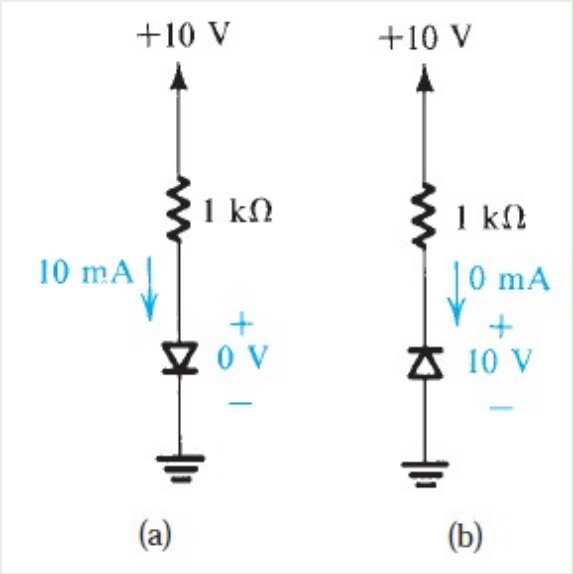


- The positive terminal of the diode is called *the anode* and the negative terminal *the cathode*, a carryover from the days of vacuum-tube diodes. The i–v characteristic of the ideal diode (conducting in one direction and not in the other) should explain the choice of its arrow like circuit symbol.

The Ideal Diode

Example

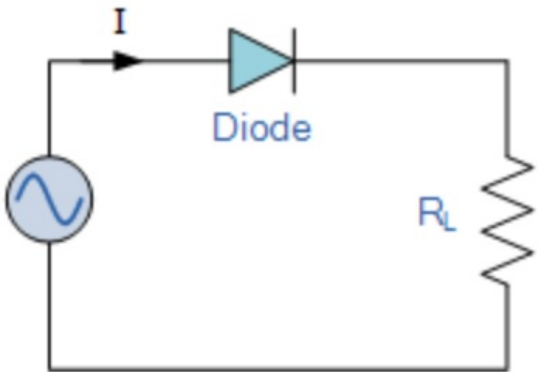
Consider the circuit illustrated below.



In the first circuit, the diode is obviously conducting. Thus its voltage drop will be zero, and the current through it will be determined by the $+10V$ supply and the $1k\Omega$ resistor as $10mA$. The diode in the second circuit is obviously cut off, and thus its current will be zero, which in turn means that the entire $10V$ supply will appear as reverse bias across the diode.

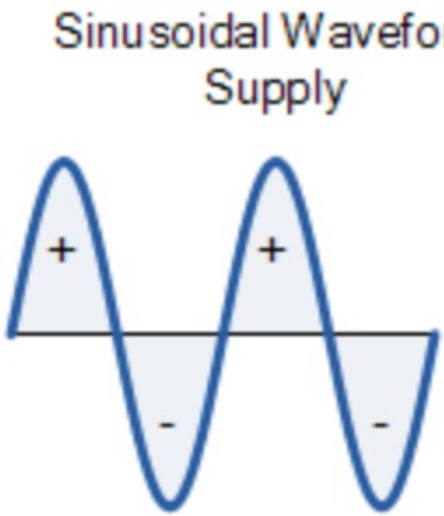
Applications of Diodes: Rectifiers

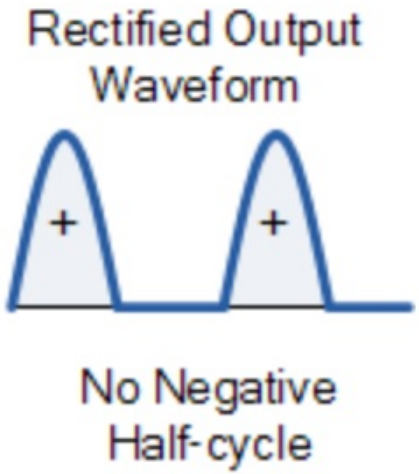
A half-wave rectifier



- During each "positive" half cycle of the AC sine wave, the diode is forward biased as the anode is positive with respect to the cathode resulting in current flowing through the diode.
- During each "negative" half cycle of the AC sinusoidal input waveform, the diode is reverse biased as the anode is negative with respect to the cathode. Therefore, NO current flows through the diode or circuit. Then in the negative half cycle of the supply, no current flows in the load resistor as no voltage appears across it so therefore, $V_{R_L} = 0$.

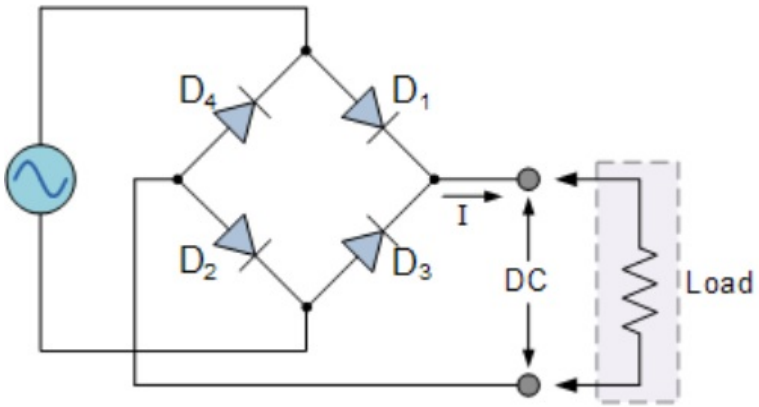
Applications of Diodes: Rectifiers





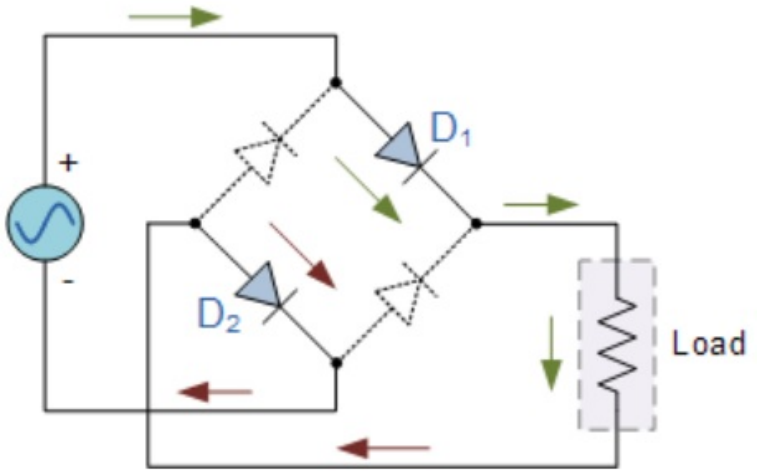
Applications of Diodes: Rectifiers

A full-wave bridge rectifier



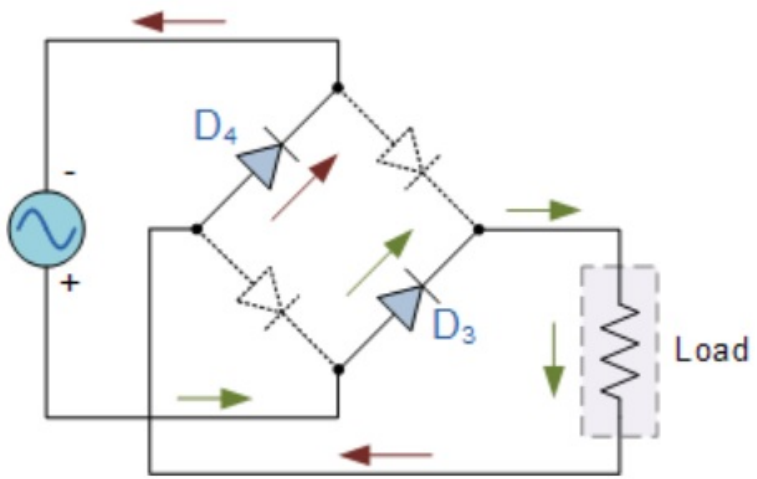
- The four diodes labelled D1 to D4 are arranged in "series pairs" with only two diodes conducting current during each half cycle.

Applications of Diodes: Rectifiers



- During the positive half cycle of the supply, diodes D1 and D2 conduct in series while diodes D3 and D4 are reverse biased and the current flows through the load as shown above.

Applications of Diodes: Rectifiers

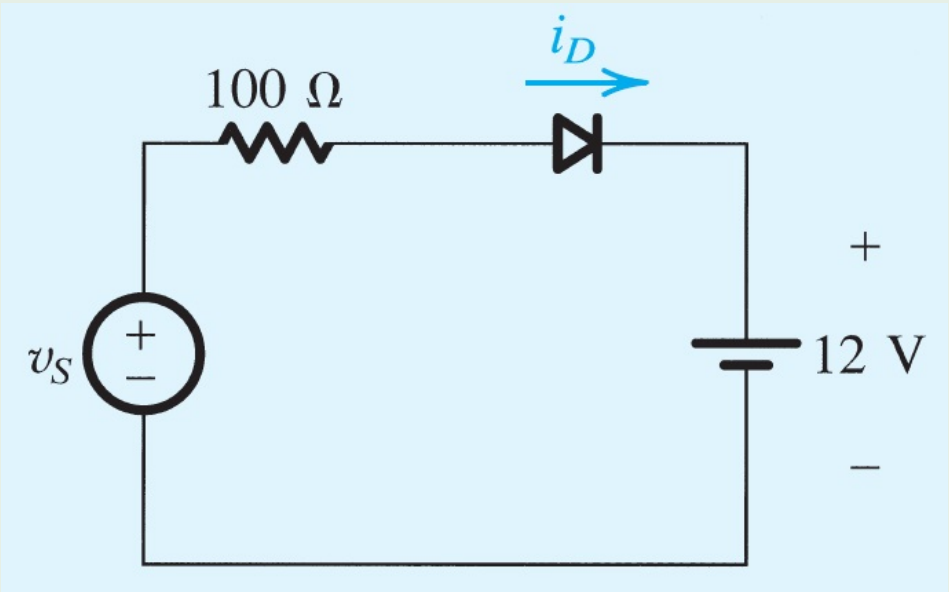


- During the negative half cycle of the supply, diodes D3 and D4 conduct in series, but diodes D1 and D2 switch "OFF" as they are now reverse biased. The current flowing through the load is the same direction as before.

Battery Charging Circuit

Example

The figure below shows a circuit for charging a $12V$ battery. If v_s is a sinusoid with $24V$ peak amplitude, draw the waveform of i_D.



Battery Charging Circuit

Answer:

